

# **Artificial Intelligence & Machine Learning - Task 4 Report**

## **Introduction**

This report presents the implementation and evaluation of classification models using the Breast Cancer dataset. The goal is to build a binary classification system and evaluate it using proper metrics beyond accuracy. Additionally, techniques for handling imbalanced data are explored.

## 1. Metric Selection

Accuracy is not sufficient for evaluating classification models, especially in imbalanced datasets. Precision measures the correctness of positive predictions, while recall measures the ability to identify actual positives. F1-score balances precision and recall, making it useful when both metrics are important. ROC curve visualizes model performance across thresholds, and AUC summarizes it into a single value. In medical diagnosis, recall is more important as missing a positive case can be critical.

## 2. Confusion Matrix Interpretation

A confusion matrix is used to evaluate classification performance by comparing predicted and actual values. It consists of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). TP: Correctly predicted positive cases. TN: Correctly predicted negative cases. FP: Incorrectly predicted positive cases. FN: Incorrectly predicted negative cases. Understanding these values helps in calculating precision, recall, and F1-score.

### **3. Handling Imbalanced Data**

Imbalanced datasets occur when one class is significantly more frequent than the other. This leads to biased models that favor the majority class. To address this issue, class weighting is applied. Logistic Regression supports `class_weight='balanced'`, which automatically adjusts weights based on class frequency. This improves model sensitivity to minority classes and enhances recall.

## 4. Model Comparison

Two models were used: Logistic Regression and Decision Tree. Logistic Regression: - Performs well on linearly separable data - Less prone to overfitting - Produces stable results Decision Tree: - Easy to interpret - Can capture complex patterns - Prone to overfitting Based on evaluation metrics, Logistic Regression showed better generalization.

## 5. Final Model Decision

After evaluating both models using confusion matrix, classification report, ROC curve, and AUC score, Logistic Regression is selected as the final model. It provides a balanced trade-off between precision and recall and performs consistently across different datasets.